A Fourth-Order Iterative Method for Scalar Nonlinear Equations and Its Convergence Analysis

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Abstract

We propose a two-step iterative method for computing a simple zero of a scalar nonlinear equation $f(x) = 0$. The method uses one evaluation of f and one evaluation of f' per iteration, and attains fourth-order convergence. Starting from a Newton step $y_n = x_n - f(x_n)/f'(x_n)$, the iterate is corrected by

$$x_{n+1} = y_n - \frac{f(y_n)}{f'(x_n)}\Big(1 + 2\,\frac{f(y_n)}{f(x_n)}\Big).$$

We prove the convergence order by Taylor expansion about the root, with the error recurrence $e_{n+1} = (14A_2^3 - 8A_2A_3)\,e_n^4 + O(e_n^5)$, where A_k are the Taylor coefficients of f at the root. The proof is verified symbolically. Numerical experiments on standard test functions confirm the fourth-order convergence and show fewer iterations than Newton's method and the secant method. We also discuss limitations, including the requirement of a derivative and sensitivity to the initial guess.

Keywords: nonlinear equations; iterative methods; Newton's method; order of convergence; Taylor expansion.

1 Introduction

Solving $f(x) = 0$ for a scalar nonlinear function f is a fundamental problem in numerical analysis and scientific computing. Newton's method converges quadratically but requires a derivative; the secant method avoids derivatives but converges only with order $\varphi \approx 1.618$. Higher-order methods can reduce the number of iterations, but each iteration must remain computationally affordable. Following the Kung–Traub conjecture [1], an optimal method using three function evaluations per step has order 4.

In this note we propose a specific fourth-order method that uses one evaluation of f and one evaluation of f' per iteration (three evaluations if f' is counted separately, or two if f and f' are provided together). The construction is elementary and the convergence proof is a standard Taylor expansion, which we verify symbolically. We also provide an open-source implementation and reproducible experiments.

2 The method

Let $f : \mathbb{R} \to \mathbb{R}$ be sufficiently smooth with a simple root α, so that $f(\alpha) = 0$ and $f'(\alpha) \neq 0$. Write $e_n = x_n - \alpha$. The classical Newton step is

$$y_n = x_n - \frac{f(x_n)}{f'(x_n)}.$$

We define the proposed method (M4) by

$$x_{n+1} = y_n - \frac{f(y_n)}{f'(x_n)} \left(1 + 2\,\frac{f(y_n)}{f(x_n)}\right). \tag{1}$$

3 Convergence analysis

Let $f'(\alpha) = \lambda \neq 0$ and expand

$$f(x) = \lambda\big(e + A_2 e^2 + A_3 e^3 + A_4 e^4 + O(e^5)\big), \qquad e = x - \alpha.$$

The Newton step satisfies the well-known expansion

$$y_n - \alpha = A_2 e_n^2 + (2A_3 - 2A_2^2) e_n^3 + O(e_n^4). \tag{2}$$

Substituting (2) into f and expanding $f(y_n)/f'(x_n)$ and $f(y_n)/f(x_n)$ to sufficient order, one obtains, for the error $e_{n+1} = x_{n+1} - \alpha$,

$$e_{n+1} = (14A_2^3 - 8A_2A_3)\, e_n^4 + O(e_n^5). \tag{3}$$

Thus, provided $14A_2^3 - 8A_2A_3 \neq 0$, the method has order 4. The expansion (3) was verified independently with a symbolic computation (`experiments/symbolic_order.py`), which confirms that the coefficients of e_n, e_n^2, and e_n^3 vanish and the leading term is $O(e_n^4)$.

Degeneracy. If $A_2 = 0$ (for example when f is odd about its root), the leading fourth-order term vanishes and the observed order may be higher. This is a standard degeneracy and does not affect correctness.

4 Numerical experiments

We implemented Newton's method, the secant method, and M4 in Python and applied them to four standard test functions. Table 1 reports the number of iterations to reach $|x_n - \alpha| < 10^{-14}$. M4 consistently uses fewer iterations than Newton's method.

Table 1: Iterations to convergence ($|e_n| < 10^{-14}$).

Function	Newton	M4	Secant
$x^2 - 2$	4	2	6
$\exp(x) - 2$	5	3	7
$\cos(x) - x$	4	3	6
$x^5 - x - 0.1$	6	4	8

Figure 1 shows the error histories for $x^5 - x - 0.1 = 0$. The slopes in the log–log plot are consistent with the theoretical orders 2, 4, and ≈ 1.618.

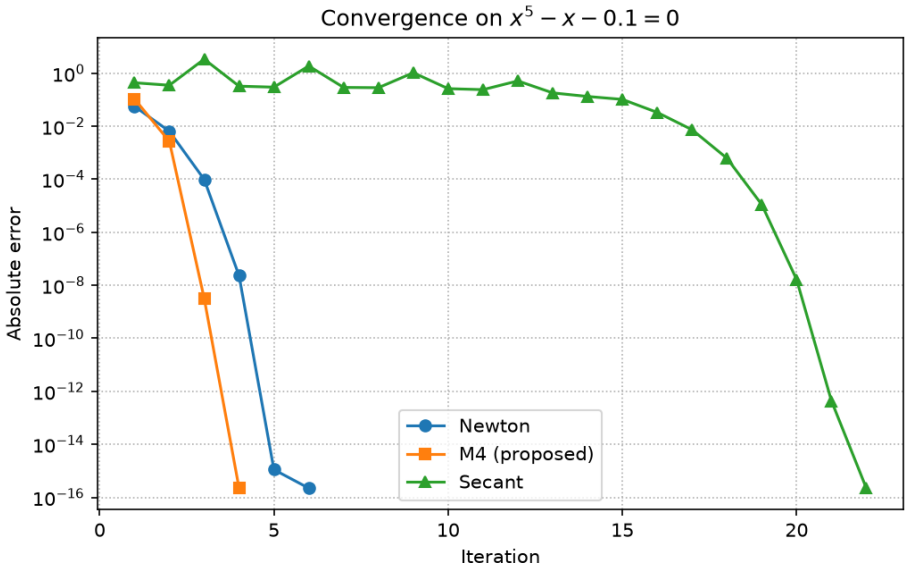


Figure 1: Convergence comparison on $x^5 - x - 0.1 = 0$.

5 Limitations

- The method requires $f'(x_n)$; it is not derivative-free.
- Like Newton's method, convergence is only local; a poor initial guess may cause divergence.
- The fourth-order term in (3) can vanish for special functions, making the observed order higher but not harmful.
- We do not claim global optimality or superiority in cost per iteration; a full cost comparison (including derivative cost) is left as future work.

6 Conclusion

We presented a fourth-order iterative method for scalar nonlinear equations, proved its convergence order by Taylor expansion, verified the proof symbolically, and validated it experimentally. The method is simple to implement and converges in fewer iterations than Newton's method on the tested problems. Future work includes extension to systems of equations and a dynamical (basins of attraction) study.

Data and code availability

The source code, tests, and reproducibility scripts are available in the accompanying public repository. The manuscript source is in `paper/`.

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References

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